StudyQuest - Business Rules Specification (English)

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1. Reward System

1.1 Study Completion Rewards

Experience Points (EXP) Calculation

```
EXP = Study Time (minutes) × 1
```

Examples:

- 30 minutes → 30 EXP
- 60 minutes → 60 EXP
- 120 minutes → 120 EXP

Coins Calculation

```
Coins = Study Time (minutes) × 5
```

Examples:

- 30 minutes → 150 Coins
- 60 minutes → 300 Coins
- 120 minutes → 600 Coins

1.2 Review Completion Rewards

Users **evaluate the difficulty on a 5-point scale** upon completing a review:

Rating	Meaning	quality value
⭐⭐⭐⭐⭐ Perfect	Remembered perfectly	5
⭐⭐⭐⭐ Easy	It was easy	4
⭐⭐⭐ Good	Normal	3

Rating	Meaning	quality value
⭐⭐ Hard	It was difficult	2
⭐ Again	Need to review again	1

Experience Points (EXP) Calculation

```
Base EXP = 5
Difficulty Bonus = (6 − quality) × 1

Total EXP = Base EXP + Difficulty Bonus
```

Examples:

Rating	quality	Calculation	EXP Earned
Perfect (5)	5	5 + (6-5)×1 = 5 + 1	**6 EXP**
Easy (4)	4	5 + (6-4)×1 = 5 + 2	**7 EXP**
Good (3)	3	5 + (6-3)×1 = 5 + 3	**8 EXP**
Hard (2)	2	5 + (6-2)×1 = 5 + 4	**9 EXP**
Again (1)	1	5 + (6-1)×1 = 5 + 5	**10 EXP**

Design Intent: Harder reviews reward more EXP (effort is rewarded)

Coins Calculation

```
Coins = 20 (fixed)
```

All review completions reward 20 coins uniformly

1.3 Level System

EXP Required for Level Up

```
EXP required for next level = Current Level × 100
```

Examples:

- Level 1→2: 100 EXP required
- Level 2→3: 200 EXP required
- Level 3→4: 300 EXP required

- Level 10→11: 1,000 EXP required

Current Level Calculation

```
Level = floor(√(Total EXP / 100)) + 1
```

Examples:

- 0 EXP → Level 1
- 150 EXP → Level 2
- 450 EXP → Level 3
- 10,000 EXP → Level 11

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2. Review Scheduling (SM2 Algorithm)

2.1 Algorithm Overview

StudyQuest uses the **SuperMemo 2 (SM2) Algorithm** to calculate optimal review intervals.

Purpose: Based on Ebbinghaus's forgetting curve, reviewing at the moment you're about to forget helps form long-term memory

2.2 Parameters

Each review item has the following parameters:

Parameter	Description	Initial Value
`interval`	Days until next review	1
`repetitions`	Number of successful reviews	0
`easeFactor`	Ease of memorization coefficient	2.5
`quality`	User rating during review	-

2.3 Calculation Logic

Step 1: Update Ease Factor

```
New EF = Old EF + (0.1 − (5 − quality) × (0.08 + (5 − quality) × 0.02))

If EF < 1.3, set to 1.3 (minimum value)
```

Examples:

quality	Calculation	Result	Meaning
5	EF + 0.10	EF increases	Longer interval
4	EF + 0.00	EF maintained	Same interval
3	EF - 0.14	EF decreases	Slightly shorter interval
2	EF - 0.32	EF significantly decreases	Shorter interval
1	EF - 0.54	EF significantly decreases	Much shorter interval

Step 2: Calculate Next Interval

```
if (quality < 3) {
    // Review failed
    interval = 1
    repetitions = 0
} else {
    // Review succeeded
    if (repetitions == 0) {
        interval = 1
    } else if (repetitions == 1) {
        interval = 6
    } else {
        interval = previous interval × easeFactor (round up)
    }
    repetitions = repetitions + 1
}
```

2.4 Review Schedule Examples

Example 1: All reviews with Good (3)

Review #	quality	repetitions	easeFactor	interval	Next Review
Initial	-	0	2.5	1	Tomorrow
1st	3	1	2.36	1	Tomorrow
2nd	3	2	2.22	6	6 days later
3rd	3	3	2.08	14	14 days later
4th	3	4	1.94	28	28 days later
5th	3	5	1.8	51	51 days later

Example 2: All reviews with Perfect (5)

Review #	quality	repetitions	easeFactor	interval	Next Review

Review #	quality	repetitions	easeFactor	interval	Next Review
Initial	-	0	2.5	1	Tomorrow
1st	5	1	2.6	1	Tomorrow
2nd	5	2	2.7	6	6 days later
3rd	5	3	2.8	17	17 days later
4th	5	4	2.9	50	50 days later
5th	5	5	3.0	150	150 days later

Example 3: Selecting Hard (2)

Review #	quality	repetitions	easeFactor	interval	Next Review
1st	3	1	2.36	1	Tomorrow
2nd	2	0	2.04	1	Tomorrow (reset)
3rd	3	1	1.9	1	Tomorrow
4th	3	2	1.76	6	6 days later

Important: If quality < 3, repetitions resets to 0, starting over from the beginning.

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3. Consecutive Study Bonus

3.1 Streak Calculation

```
if (Studied today && Studied yesterday) {
    streak = streak + 1
} else if (Studied today && Did not study yesterday && Before day before
yesterday) {
    streak = 1 (reset)
}
```

Rules:

- Streak increases when studying every day
- Resets if even one day is missed
- Maintained if studying within the same day

3.2 Streak Bonuses

Milestone Rewards

Consecutive Days	Bonus EXP	Bonus Coins

Consecutive Days	Bonus EXP	Bonus Coins
7 days	100	500
14 days	300	1,500
30 days	1,000	5,000
60 days	2,500	12,500
100 days	5,000	25,000

Daily Bonus (Cafe Effect)

```
if (streak >= 3) {
    Bonus EXP = Base EXP × 0.1 × (1 + CafeLevel × 0.05)
    Bonus Coins = Base Coins × 0.1 × (1 + CafeLevel × 0.05)
}
```

Example (30 minutes study, Cafe level 3, streak=5 days):

- Base: 30 EXP, 150 Coins
- Bonus: 30×0.1×1.15 = 3.45 → 3 EXP
- Bonus: 150×0.1×1.15 = 17.25 → 17 Coins
- Total: 33 EXP, 167 Coins